

ID Scanner Application

The **ID Scanner Application** is a user-friendly tool built with **Streamlit** that leverages **Optical Character Recognition (OCR)** technology to extract and display text from uploaded ID images. It is designed to simplify the process of extracting key information from IDs, such as organization names and employee details.

Features

- **Image Upload:** Supports multiple image formats, including `.jpg`, `.png`, `.jpeg`, and `.webp`.
 - **OCR Processing:** Uses Tesseract OCR to extract text from uploaded images.
 - **Organized Output:** Displays extracted text in a structured format:
 - Organization Name
 - Employee Name
 - Additional details from the ID
 - **Interactive UI:** Built with Streamlit for a clean and responsive user interface.
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Requirements

- **Python:** Version 3.7 or higher
 - **Libraries:**
 - `numpy`
 - `streamlit`
 - `pytesseract`
 - `opencv-python`
 - `Pillow`
 - **Tesseract OCR:**
 - Installable from [Tesseract GitHub Wiki](#).
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Installation

Follow these steps to set up the project on your local machine:

1. Clone the Repository:

```
git clone <repository-url>
cd ID_SCAN
```

2. Install Dependencies:

Install the required Python libraries using `pip`:

```
pip install -r requirements.txt
```

3. Install Tesseract OCR:

- Download and install Tesseract OCR from [Tesseract GitHub Wiki](#).
- Update the `pytesseract.pytesseract.tesseract_cmd` path in the code to match your Tesseract installation directory.

Usage

1. Run the Application:

Launch the Streamlit app by running the following command:

```
streamlit run ID_SCAN.py
```

2. Upload an Image:

Use the file uploader in the web interface to upload an image of an ID.

3. View Extracted Text:

The extracted text will be displayed in a structured format on the page.

File Structure

```
ID_SCAN/  
├── ID_SCAN.ipynb      # Jupyter Notebook for development  
├── ID_SCAN.py         # Main Python script for the Streamlit app  
├── requirements.txt   # List of required Python libraries  
└── README.md         # Project documentation
```

Notes

- Ensure that **Tesseract OCR** is installed and properly configured on your system.
- The application assumes a specific format for the ID text. Adjustments may be required for different ID layouts.
- For best results, use high-quality images with clear text.

License

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Acknowledgments

- [Tesseract OCR](#) for its powerful OCR capabilities.
- [Streamlit](#) for providing an intuitive framework for building interactive web applications.

Future Enhancements

- Add support for additional languages in OCR.
- Improve text parsing for better accuracy with various ID formats.
- Implement a database to store extracted information for future reference.

Feel free to contribute to this project by submitting issues or pull requests. Happy coding! 